

CLASS 10 · CBSE · NCERT MATHEMATICS · PYQ SOLUTIONS

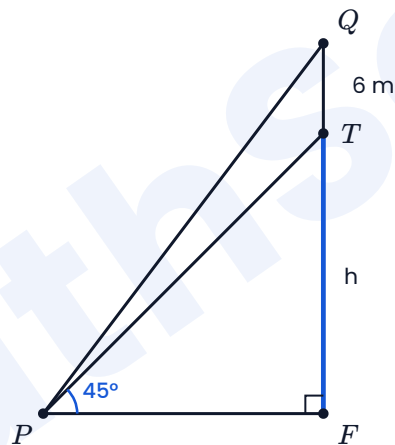
Chapter 9: Some Applications of Trigonometry

Step-by-step solutions to the previous year questions

The question numbers match the PYQ question sheet.

1 Two right-angled triangles

1. A pole 6 m high is fixed on the top of a tower. The angle of elevation of the top of the pole observed from a point P on the ground is 60° and the angle of depression of the point P from the top of the tower is 45° . Find the height of the tower and the distance of point P from the foot of the tower. (Use $\sqrt{3} = 1.73$) **CBSE 2024, 5 marks**



FT is the tower and TQ is the pole

SOLUTION

Let the height of the tower be h m and the distance of P from its foot be d m.

The angle of depression of P from the top of the tower is 45° , so the angle of elevation of the top of the tower from P is 45° : $\tan 45^\circ = \frac{h}{d}$, so $d = h$.

For the top of the pole: $\tan 60^\circ = \frac{h+6}{d}$, so $\sqrt{3}h = h + 6$.

$$h = \frac{6}{\sqrt{3}-1} = \frac{6(\sqrt{3}+1)}{2} = 3(\sqrt{3}+1) = 3 \times 2.73 = 8.19$$

Answer: the tower is 8.19 m high and P is 8.19 m from its foot.